

4. Hausdorff measures and dimension

In this section we introduce Hausdorff measures and dimension for measuring the metric size of quite general sets. They will be one of the basic means for studying geometric properties of sets and expressing results that these studies lead to. Hausdorff measures also provide a fruitful source for getting examples to which several later results on general measures apply. The basic definitions and first results on Hausdorff measures and dimension are due to Carathéodory [1] and Hausdorff [1]. We shall start with a more general construction, called Carathéodory's construction. It will yield also many other measures some of which will be briefly presented in the next chapter.

Carathéodory's construction

4.1. Let X be a metric space, $\mathcal{F}$ a family of subsets of X and ζ a non-negative function on $\mathcal{F}$. We make the following two assumptions.

- (1) For every $\delta > 0$ there are $E_1, E_2, \dots \in \mathcal{F}$ such that $X = \bigcup_{i=1}^{\infty} E_i$ and $d(E_i) \leq \delta$.
- (2) For every $\delta > 0$ there is $E \in \mathcal{F}$ such that $\zeta(E) \leq \delta$ and $d(E) \leq \delta$.

For $0 < \delta \leq \infty$ and $A \subset X$ we define

$$\psi_{\delta}(A) = \inf \left\{ \sum_{i=1}^{\infty} \zeta(E_i) : A \subset \bigcup_{i=1}^{\infty} E_i, d(E_i) \leq \delta, E_i \in \mathcal{F} \right\}.$$

Assumption (1) was only introduced to guarantee that such coverings always exist. The role of (2) is to have $\psi_{\delta}(\emptyset) = 0$. It also allows us to use coverings $\{E_i\}_{i \in I}$ with I finite or countable without changing the value of $\psi_{\delta}(A)$.

It is easy to see that ψ_{δ} is monotonic and subadditive so that it is a measure. Usually it is highly non-additive and not a Borel measure (see Exercise 1). Evidently,

$$\psi_{\delta}(A) \leq \psi_{\varepsilon}(A) \quad \text{whenever } 0 < \varepsilon < \delta \leq \infty.$$

Hence we can define $\psi = \psi(\mathcal{F}, \zeta)$ by

$$\psi(A) = \lim_{\delta \downarrow 0} \psi_{\delta}(A) = \sup_{\delta > 0} \psi_{\delta}(A) \quad \text{for } A \subset X.$$

The measure-theoretic behaviour of ψ is much better than that of ψ_{δ} .

4.2. Theorem.

- (1) ψ is a Borel measure.
 (2) If the members of $\mathcal{F}$ are Borel sets, ψ is Borel regular.

Proof. (1) The proof that ψ is a measure is straightforward and left to the reader. To show that ψ is a Borel measure, we verify the condition of Theorem 1.7. Let $A, B \subset X$ with $d(A, B) > 0$. Choose δ with $0 < \delta < d(A, B)/2$. If the sets $E_1, E_2, \dots \in \mathcal{F}$ cover $A \cup B$ and satisfy $d(E_i) \leq \delta$, then none of them can meet both A and B . Hence

$$\begin{aligned} \sum_i \zeta(E_i) &\geq \sum_{A \cap E_i \neq \emptyset} \zeta(E_i) + \sum_{B \cap E_i \neq \emptyset} \zeta(E_i) \\ &\geq \psi_\delta(A) + \psi_\delta(B). \end{aligned}$$

Taking the infimum over all such coverings we have $\psi_\delta(A \cup B) \geq \psi_\delta(A) + \psi_\delta(B)$. But the opposite inequality holds also as ψ_δ is a measure, and so $\psi_\delta(A \cup B) = \psi_\delta(A) + \psi_\delta(B)$. Letting $\delta \downarrow 0$, we obtain $\psi(A \cup B) = \psi(A) + \psi(B)$ as required.

(2) If $A \subset X$, choose for every $i = 1, 2, \dots$ sets $E_{i,1}, E_{i,2}, \dots \in \mathcal{F}$ such that

$$\begin{aligned} A &\subset \bigcup_j E_{i,j}, \quad d(E_{i,j}) \leq 1/i \quad \text{and} \\ \sum_j \zeta(E_{i,j}) &\leq \psi_{1/i}(A) + 1/i. \end{aligned}$$

Then $B = \bigcap_i \bigcup_j E_{i,j}$ is a Borel set such that $A \subset B$ and $\psi(A) = \psi(B)$. Thus ψ is Borel regular. $\square$

Hausdorff measures

4.3. Let X be separable, $0 \leq s < \infty$, and choose

$$\begin{aligned} \mathcal{F} &= \{E : E \subset X\}, \\ \zeta(E) &= \zeta_s(E) = d(E)^s \end{aligned}$$

with the interpretations $0^0 = 1$ and $d(\emptyset)^s = 0$. The resulting measure ψ is called the s -dimensional *Hausdorff measure* and denoted by $\mathcal{H}^s$. So

$$\mathcal{H}^s(A) = \lim_{\delta \downarrow 0} \mathcal{H}_\delta^s(A)$$

where

$$\mathcal{H}_\delta^s(A) = \inf \left\{ \sum_i d(E_i)^s : A \subset \bigcup_i E_i, d(E_i) \leq \delta \right\}.$$

The integral dimensional Hausdorff measures play a special role. Let us start from $s = 0$. It is easy to see that $\mathcal{H}^0$ is the counting measure:

$$\mathcal{H}^0(A) = \text{card } A = \text{the number of points in } A.$$

Next, for $s = 1$, $\mathcal{H}^1$ also has a concrete interpretation as a generalized length measure. In particular, for a rectifiable curve Γ in $\mathbf{R}^n$, $\mathcal{H}^1(\Gamma)$ can be shown to equal the length of Γ . (If the length is defined in some other reasonable way; of course, $\mathcal{H}^1(\Gamma)$ can also be taken as the definition of the length of Γ .) For unrectifiable curves Γ , $\mathcal{H}^1(\Gamma) = \infty$. More generally, if m is an integer, $1 \leq m < n$, and M is a sufficiently regular m -dimensional surface in $\mathbf{R}^n$ (for example, C^1 submanifold), then the restriction $\mathcal{H}^m \llcorner M$ gives a constant multiple of the surface measure on M . This follows for example from the area formula, see Federer [3, 3.2.3] or Evans and Gariepy [1, § 3.3].

For $s = n$ in $\mathbf{R}^n$,

$$(1) \quad \mathcal{H}^n = 2^n \alpha(n)^{-1} \mathcal{L}^n,$$

whence

$$(2) \quad \mathcal{H}^n(B(x, r)) = (2r)^n \quad \text{for } x \in \mathbf{R}^n, 0 < r < \infty.$$

Often one normalizes Hausdorff measures (as in Federer [3] and L. Simon [1]) so that $\mathcal{H}^n$ will equal $\mathcal{L}^n$, but since we shall not usually be interested in the exact values of Hausdorff measures, we use the simpler definition. The proof of the equality (1) is rather complicated and based on the so-called isodiametric inequality

$$\mathcal{L}^n(A) \leq 2^{-n} \alpha(n) d(A)^n \quad \text{for } A \subset \mathbf{R}^n,$$

see Federer [3, 2.10.33] or Evans and Gariepy [1, § 2.2]. But to see that $\mathcal{H}^n = c\mathcal{L}^n$ with some positive and finite constant is much easier. All we have to do is to verify that both $\mathcal{H}^n$ and $\mathcal{L}^n$ are uniformly distributed measures and use Theorem 3.4. (That $\mathcal{H}^n$ is Borel regular will be noted in 4.5.) We shall use the formulas (1) and (2) many times, but almost always the weaker information that they hold with some unspecified constants would suffice. Only in the proof of Theorem 16.2 do we rely on the precise form of (2).

For any $s > n$, $\mathcal{H}^s$ in $\mathbf{R}^n$ is uninteresting since $\mathcal{H}^s(\mathbf{R}^n) = 0$ (see Theorem 4.7).

Hausdorff measures behave nicely under translations and dilations in $\mathbf{R}^n$: for $A \subset \mathbf{R}^n$, $a \in \mathbf{R}^n$, $0 < t < \infty$,

$$\begin{aligned}\mathcal{H}^s(A + a) &= \mathcal{H}^s(A) \quad \text{where } A + a = \{x + a : x \in A\}, \\ \mathcal{H}^s(tA) &= t^s \mathcal{H}^s(A) \quad \text{where } tA = \{tx : x \in A\}.\end{aligned}$$

These are readily verified from the definition. In particular,

$$\mathcal{H}^s(B(x, r)) = c(s, n) r^s \quad \text{for } x \in \mathbf{R}^n, \ 0 < r < \infty.$$

But, as follows from Theorem 4.7, $c(s, n)$ is positive and finite only when $s = n$; for $s > n$, $c(s, n) = 0$, for $s < n$, $c(s, n) = \infty$. Thus only $\mathcal{H}^n$ is uniformly distributed in $\mathbf{R}^n$. To prove that $0 < c(n, n) < \infty$, one can use any of the standard proofs for the fact that the unit ball (or cube) has positive and finite Lebesgue measure.

We shall now derive some simple properties of Hausdorff measures in a general separable metric space X .

4.4. Theorem. Let $0 \leq s < n$ and $\zeta(E) = d(E)^s$ for $E \subset X$. If

- (1) $\mathcal{F} = \{F \subset X : F \text{ is closed}\}$ or
- (2) $\mathcal{F} = \{U \subset X : U \text{ is open}\}$ or
- (3) $X = \mathbf{R}^n$ and $\mathcal{F} = \{K \subset \mathbf{R}^n : K \text{ is convex}\}$,

then $\psi(\mathcal{F}, \zeta) = \mathcal{H}^s$.

The first and last statement follow from the fact that the closure and convex hull of a set E have the same diameter as E . The second statement holds since for any $\varepsilon > 0$, $\{x : d(x, E) < \varepsilon\}$ is open and has diameter at most $d(E) + 2\varepsilon$. We leave the details as an exercise.

Recalling Theorem 4.2 (2) we have

4.5. Corollary. $\mathcal{H}^s$ is Borel regular.

Notice that usually $\mathcal{H}^s$ is not a Radon measure since it need not be locally finite. For example, if $s < n$ every non-empty open set in $\mathbf{R}^n$ has non- σ -finite $\mathcal{H}^s$ measure. But taking any $\mathcal{H}^s$ measurable set A in $\mathbf{R}^n$ with $\mathcal{H}^s(A) < \infty$, the restriction $\mathcal{H}^s \llcorner A$ is a Radon measure by Theorem 1.9 (2) and Corollary 1.11.

Often one is only interested in knowing which sets have $\mathcal{H}^s$ measure zero. For this it is enough to use any of the approximating measures $\mathcal{H}_\delta^s$, for example $\mathcal{H}_\infty^s$; in fact we don't really need any measure at all.

4.6. Lemma. Let $A \subset X$, $0 \leq s < \infty$ and $0 < \delta \leq \infty$. Then the following conditions are equivalent:

- (1) $\mathcal{H}^s(A) = 0$.
- (2) $\mathcal{H}_\delta^s(A) = 0$.
- (3) $\forall \varepsilon > 0 \exists E_1, E_2, \dots \subset X$ such that

$$A \subset \bigcup_i E_i \quad \text{and} \quad \sum_i d(E_i)^s < \varepsilon.$$

The proof is left as an exercise.

We shall now compare measures $\mathcal{H}^s$ with each other.

4.7. Theorem. For $0 \leq s < t < \infty$ and $A \subset X$,

- (1) $\mathcal{H}^s(A) < \infty$ implies $\mathcal{H}^t(A) = 0$,
- (2) $\mathcal{H}^t(A) > 0$ implies $\mathcal{H}^s(A) = \infty$.

Proof. To prove (1), let $A \subset \bigcup_i E_i$ with $d(E_i) \leq \delta$ and $\sum_i d(E_i)^s \leq \mathcal{H}_\delta^s(A) + 1$. Then

$$\mathcal{H}_\delta^t(A) \leq \sum_i d(E_i)^t \leq \delta^{t-s} \sum_i d(E_i)^s \leq \delta^{t-s} (\mathcal{H}_\delta^s(A) + 1),$$

which gives (1) as $\delta \downarrow 0$.

(2) is really only a restatement of (1). But we have emphasized this simple theorem by doublestating it, because it leads to one of the most fundamental concepts of this book, the Hausdorff dimension.

Hausdorff dimension

According to Theorem 4.7 we may define

4.8. Definition. The Hausdorff dimension of a set $A \subset X$ is

$$\begin{aligned} \dim A &= \sup\{s : \mathcal{H}^s(A) > 0\} = \sup\{s : \mathcal{H}^s(A) = \infty\} \\ &= \inf\{t : \mathcal{H}^t(A) < \infty\} = \inf\{t : \mathcal{H}^t(A) = 0\}. \end{aligned}$$

(Sometimes some of these sets may be empty, but we leave the obvious interpretations to the reader.)

Clearly the Hausdorff dimension has the natural properties of monotonicity and stability with respect to countable unions:

$$\begin{aligned} \dim A &\leq \dim B && \text{for } A \subset B \subset X, \\ \dim \bigcup_{i=1}^{\infty} A_i &= \sup_i \dim A_i && \text{for } A_i \subset X, \ i = 1, 2, \dots \end{aligned}$$

To state the definition in other words, $\dim A$ is the unique number (it may be ∞ in some metric spaces) for which

$$\begin{aligned} s < \dim A &\text{ implies } \mathcal{H}^s(A) = \infty, \\ t > \dim A &\text{ implies } \mathcal{H}^t(A) = 0. \end{aligned}$$

At the borderline case $s = \dim A$ we cannot have any general non-trivial information about the value $\mathcal{H}^s(A)$; all three cases $\mathcal{H}^s(A) = 0$, $0 < \mathcal{H}^s(A) < \infty$, $\mathcal{H}^s(A) = \infty$ are possible. But if for some given A we can find s such that $0 < \mathcal{H}^s(A) < \infty$, then s must equal $\dim A$.

Since $\mathbf{R}^n$ has infinite but σ -finite $\mathcal{H}^n$ measure, it follows that

$$\dim \mathbf{R}^n = n.$$

Hence $0 \leq \dim A \leq n$ for all $A \subset \mathbf{R}^n$. We shall soon see that for all $s \in [0, n]$, $\dim A = s$ for some subset A of $\mathbf{R}^n$.

To find the Hausdorff dimension or to estimate the Hausdorff measures of a given set, it is always possible and often advantageous to use coverings with some simpler sets like balls or, in $\mathbf{R}^n$, dyadic cubes. This is easy to see and we shall return to it in the next chapter.

Recalling Lemma 4.6 (3) we observe that we do not really need Hausdorff measures to define Hausdorff dimension.

Generalized Hausdorff measures

4.9. Although the Hausdorff dimension measures the metric size of any subset of our metric space, the values of the Hausdorff measures often do not give much extra information. This is so since there may be no value s for which the set has positive and finite $\mathcal{H}^s$ measure. But often replacing $\zeta_s(E) = d(E)^s$ by some other function of the diameter, one can find measures measuring the given set in a more delicate manner.

Let $h: [0, \infty) \rightarrow [0, \infty)$ be a non-decreasing function with $h(0) = 0$. We take again

$$\mathcal{F} = \{E : E \subset X\} \quad \text{and} \quad \zeta(E) = h(d(E))$$

(with $d(\emptyset) = 0$). Then the corresponding measure $\psi(\mathcal{F}, \zeta) = \Lambda_h$ is called the Hausdorff h measure. Of course, $\Lambda_h = \mathcal{H}^s$ when $h(t) = t^s$.

There are many cases where some other h than t^s is more useful and natural. Among the most important are sets related to Brownian motion in $\mathbf{R}^n$. For example, the trajectories of the Brownian motion in $\mathbf{R}^n$ have positive and σ -finite Λ_h measure almost surely with (for small t)

$$\begin{aligned} h(t) &= t^2 \log \log t^{-1} && \text{in the case } n \geq 3, \text{ and} \\ h(t) &= t^2 \log t^{-1} \log \log \log t^{-1} && \text{in the case } n = 2. \end{aligned}$$

In particular, their dimension is 2 almost surely. For more on this topic, see Falconer [4], [16], Kahane [3], Adler [1] and Taylor [2]. The generalized Hausdorff measures are also useful in many questions of complex analysis, see e.g. Makarov [1] and Pommerenke [1]. Their scaling properties were studied by Mauldin and Williams [4]. Some Hausdorff-type measures in big metric spaces were examined by Johnson and Rogers [1].

We have now introduced measures for measuring the size of very general sets. It is time to look at some examples with which Hausdorff measures are convenient and useful. We begin with the most classical.

Cantor sets

4.10. Cantor sets in $\mathbf{R}^1$. Let $0 < \lambda < 1/2$. Denote $I_{0,1} = [0, 1]$, and let $I_{1,1}$ and $I_{1,2}$ be the intervals $[0, \lambda]$ and $[1 - \lambda, 1]$, respectively. We continue this process of selecting two subintervals of each already given interval. If we have defined intervals $I_{k-1,1}, \dots, I_{k-1,2^{k-1}}$, we define $I_{k,1}, \dots, I_{k,2^k}$ by deleting from the middle of each $I_{k-1,j}$ an interval of length $(1 - 2\lambda) d(I_{k-1,j}) = (1 - 2\lambda) \lambda^{k-1}$. All the intervals $I_{k,j}$ thus obtained have length λ^k .

We define a kind of limit set of this construction by

$$C(\lambda) = \bigcap_{k=0}^{\infty} \bigcup_{j=1}^{2^k} I_{k,j}.$$

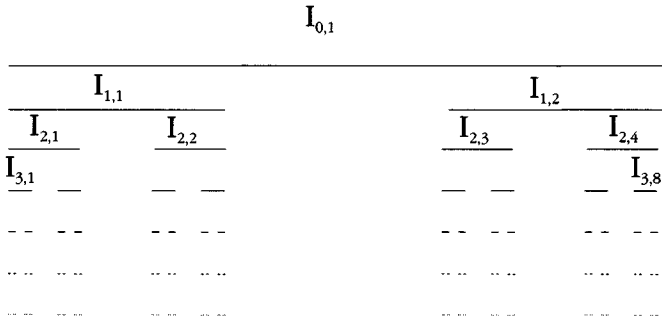


Figure 4.1. A Cantor set.

Then $C(\lambda)$ is an uncountable compact set without interior points and with zero Lebesgue measure. The most commonly used case is the Cantor middle-third set $C(1/3)$, see Figure 4.1.

We shall now study the Hausdorff measures and dimension of $C(\lambda)$. As usual, it is much simpler to find upper bounds than lower bounds for the Hausdorff measures. This is due to the definition: a judiciously chosen covering will give an upper estimate, but a lower estimate requires finding an infimum over arbitrary coverings.

For every $k = 1, 2, \dots$, $C(\lambda) \subset \bigcup_j I_{k,j}$, and so

$$\mathcal{H}_{\lambda^k}^s(C(\lambda)) \leq \sum_{j=1}^{2^k} d(I_{k,j})^s = 2^k \lambda^{ks} = (2\lambda^s)^k.$$

In order for this upper bound to be useful, it should stay bounded as $k \rightarrow \infty$. The smallest value of s for which this happens is given by $2\lambda^s = 1$, that is,

$$s = \log 2 / \log(1/\lambda).$$

For this choice we have

$$\mathcal{H}^s(C(\lambda)) = \lim_{k \rightarrow \infty} \mathcal{H}_{\lambda^k}^s(C(\lambda)) \leq 1.$$

Thus $\dim C(\lambda) \leq s$.

Next we shall show

$$(1) \quad \mathcal{H}^s(C(\lambda)) \geq 1/4,$$

which will give

$$\dim C(\lambda) = \log 2 / \log(1/\lambda).$$

To prove (1), it suffices to show that

$$(2) \quad \sum_j d(I_j)^s \geq 1/4$$

whenever open intervals $I_1, I_2, \dots$ cover $C(\lambda)$. Since $C(\lambda)$ is compact, finitely many I_j 's cover $C(\lambda)$ so that we may assume that there were only $I_1, \dots, I_n$ to begin with. Since $C(\lambda)$ has no interior points, we can, making I_j slightly larger if necessary, assume that the end-points of each I_j are outside $C(\lambda)$. Then there is $\delta > 0$ such that the distance from all these end-points to $C(\lambda)$ is at least δ . Choosing k so large that $\delta > \lambda^k = d(I_{k,i})$, it follows that every interval $I_{k,i}$ is contained in some I_j .

We shall now show that for any open interval I and any fixed ℓ ,

$$(3) \quad \sum_{I_{\ell,i} \subset I} d(I_{\ell,i})^s \leq 4d(I)^s.$$

This gives (2), since

$$4 \sum_j d(I_j)^s \geq \sum_j \sum_{I_{k,i} \subset I_j} d(I_{k,i})^s \geq \sum_{i=1}^{2^k} d(I_{k,i})^s = 1.$$

To verify (3), suppose there are some intervals $I_{\ell,i}$ inside I and let n be the smallest integer for which I contains some $I_{n,i}$. Then $n \leq \ell$. Let $I_{n,j_1}, \dots, I_{n,j_p}$ be all the n -th generation intervals which meet I . Then $p \leq 4$, since otherwise I would contain some $I_{n-1,i}$. Thus

$$4d(I)^s \geq \sum_{m=1}^p d(I_{n,j_m})^s = \sum_{m=1}^p \sum_{I_{\ell,i} \subset I_{n,j_m}} d(I_{\ell,i})^s \geq \sum_{I_{\ell,i} \subset I} d(I_{\ell,i})^s.$$

Actually it is not hard to show that (2) can be improved to $\sum d(I_j)^s \geq 1$, which gives the precise value $\mathcal{H}^s(C(\lambda)) = 1$, see Falconer [4, Theorem 1.14]. However, the above argument can be generalized to many situations where the exact value of the measure is practically impossible to compute. Marion [1]–[2] calculated the exact value of the Hausdorff measure for a large class of self-similar sets.

Note that $\dim C(\lambda)$ measures the sizes of the Cantor sets $C(\lambda)$ in a natural way: when λ increases, the sizes of the deleted holes decrease and the sets $C(\lambda)$ become larger, and also $\dim C(\lambda)$ increases. Notice also that when λ runs from 0 to $1/2$, $\dim C(\lambda)$ takes all the values between 0 and 1.

4.11. Generalized Cantor sets in $\mathbf{R}^1$. Instead of keeping constant the ratios of the lengths of the intervals in every two successive stages of the construction, we can vary it in the following way. Let $T = (\lambda_i)$ be a sequence of numbers in the open interval $(0, 1/2)$. We construct a set $C(T)$ otherwise as above, but take the intervals $I_{k,j}$ to have length $\lambda_k d(I_{k-1,i})$. Then for every k we get 2^k intervals $I_{k,j}$ of length

$$s_k = \lambda_1 \cdots \lambda_k.$$

Let $h: [0, \infty) \rightarrow [0, \infty)$ be a continuous increasing function such that

$$(1) \quad h(s_k) = 2^{-k}.$$

Then by the above argument

$$1/4 \leq \Lambda_h(C(T)) \leq 1.$$

Conversely, we can start from any continuous increasing function $h: [0, \infty) \rightarrow [0, \infty)$ such that $h(0) = 0$ and $h(2r) < 2h(r)$ for $0 < r < \infty$, and inductively select $\lambda_1, \lambda_2, \dots$ such that (1) is valid. Thus for any such h there is a compact set $C_h \subset \mathbf{R}^1$ such that $0 < \Lambda_h(C_h) < \infty$. Choosing $h(r) = r^s \log(1/r)$ for small values of r , where $0 < s \leq 1$, we have $\dim C_h = s$ and $\mathcal{H}^s(C_h) = 0$. On the other hand, choosing $h(r) = r^s / \log(1/r)$ for small r , where $0 \leq s < 1$, C_h has non- σ -finite $\mathcal{H}^s$ measure and dimension s . In particular, the extreme cases $s = 1$ and $s = 0$ give a set of the dimension 1 with zero Lebesgue measure and an uncountable set of dimension zero.

4.12. Cantor sets in $\mathbf{R}^n$. We can use the same ideas as above to construct Cantor-type sets in $\mathbf{R}^n$ having a given Hausdorff dimension s . We can start from a ball, cube, rectangle etc. and at each stage of the construction select similar geometric figures inside the previous ones. One can then often use the following proposition.

Suppose for $k = 1, 2, \dots$ we have compact sets $E_{i_1, \dots, i_k}$, $i_j = 1, \dots, m_j$, such that

$$\begin{aligned} E_{i_1, \dots, i_k, i_{k+1}} &\subset E_{i_1, \dots, i_k}, \\ d_k = \max_{i_1 \cdots i_k} d(E_{i_1, \dots, i_k}) &\rightarrow 0 \quad \text{as } k \rightarrow \infty, \\ \sum_{j=1}^{m_{k+1}} d(E_{i_1, \dots, i_k, j})^s &= d(E_{i_1, \dots, i_k})^s, \\ \sum_{B \cap E_{i_1, \dots, i_k} \neq \emptyset} d(E_{i_1, \dots, i_k})^s &\leq cd(B)^s \end{aligned}$$

for any ball B with $d(B) \geq d_k$, where c is a positive constant. Then

$$0 < \mathcal{H}^s \left(\bigcap_{k=1}^{\infty} \bigcup_{i_1 \dots i_k} E_{i_1, \dots, i_k} \right) < \infty.$$

We leave the proof as an exercise. A related more general result is given by Martin and Mattila [1]. Notice that the above conditions are satisfied for example in the following situation: select all the sets $E_{i_1, \dots, i_k}$ to be balls of radius r_k . Choose the balls $E_{i_1, \dots, i_k, j}$ fairly uniformly distributed inside $E_{i_1, \dots, i_k}$ and so that $m_{k+1} r_{k+1}^s = r_k^s$. If r_k tends to zero very rapidly (or equivalently, m_k grows very rapidly), the diameter $2r_{k+1}$ of $E_{i_1, \dots, i_k, i}$ is much smaller for large k than the distance from $E_{i_1, \dots, i_k, i}$ to the nearest neighbour $E_{i_1, \dots, i_k, j}$; this distance is of magnitude $r_k^{1-s/n} r_{k+1}^{s/n}$. Hence sets with large Hausdorff dimension (even equal to n) can look extremely porous at arbitrarily small scales, cf. Figure 4.2.

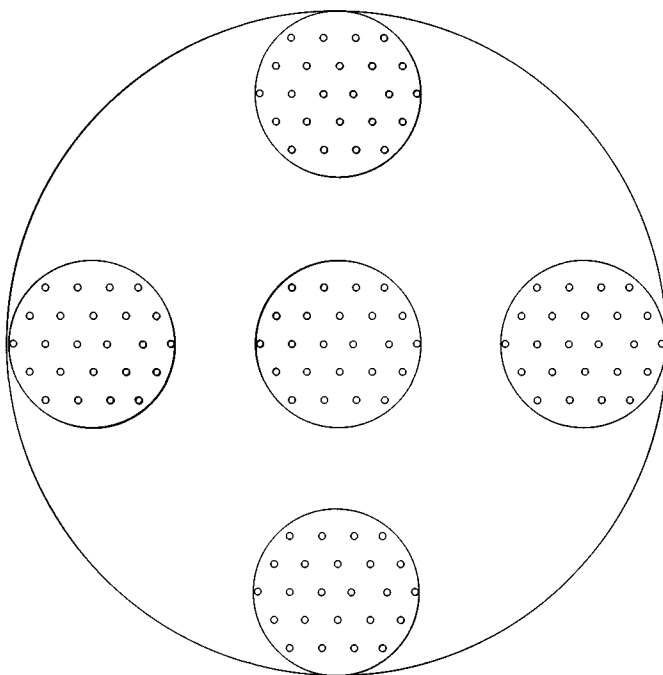


Figure 4.2. A very porous Cantor set.

Formulas for the Hausdorff dimension of Cantor sets in $\mathbf{R}^n$ constructed by means of balls were derived by Beardon [1] and by Hata [2].

Self-similar and related sets

4.13. Self-similar sets. Roughly speaking, a subset of $\mathbf{R}^n$ is self-similar if it can be split into parts which are geometrically similar to the whole set. The Cantor sets $C(\lambda)$ in 4.10 are simple examples. If the parts $C(\lambda) \cap [0, \lambda]$ and $C(\lambda) \cap [1 - \lambda, 1]$ are magnified in ratio $1/\lambda$ we get (a translate of) the original Cantor set. We shall briefly describe parts of the more general elegant theory of Hutchinson. For more details see Hutchinson [1], Falconer [4] or [16].

The self-similarity of $C(\lambda)$ above can be expressed by the formula

$$C(\lambda) = S_1(C(\lambda)) \cup S_2(C(\lambda))$$

where the similarity maps $S_1, S_2: \mathbf{R} \rightarrow \mathbf{R}$ are defined by $S_1(x) = \lambda x$, $S_2(x) = \lambda x + 1 - \lambda$. Another standard example is von Koch's "snowflake" curve, see Figure 4.3. In the construction one replaces at each stage a segment of length d by four segments of length $d/3$ as in the figure. The von Koch curve K is a limit of the polygonal curves thus obtained. It is a non-rectifiable curve having tangents at none of its points. It can also be presented in terms of similarity maps S_i in the form

$$K = S_1K \cup S_2K \cup S_3K \cup S_4K.$$

Here $S_1, \dots, S_4$ are the orientation-preserving similarities of ratios $1/3$ of the plane which map the first initial segment onto the four next ones.

We now state the basic ideas of Hutchinson's general theory. A mapping $S: \mathbf{R}^n \rightarrow \mathbf{R}^n$ is called a *similitude* if there is r , $0 < r < 1$, such that

$$|S(x) - S(y)| = r|x - y| \quad \text{for } x, y \in \mathbf{R}^n.$$

Similitudes are exactly those maps S which can be written as

$$S(x) = rg(x) + z, \quad x \in \mathbf{R}^n,$$

for some $g \in O(n)$, $z \in \mathbf{R}^n$ and $0 < r < 1$. Suppose $\mathcal{S} = \{S_1, \dots, S_N\}$, $N \geq 2$, is a finite sequence of similitudes with contraction ratios $r_1, \dots, r_N$. We say that a non-empty compact set K is *invariant* under $\mathcal{S}$ if

$$K = \bigcup_{i=1}^N S_i K.$$

Then for any such $\mathcal{S}$ there exists a unique invariant compact set. A quick way to prove this is to use the fact that the family of all non-empty

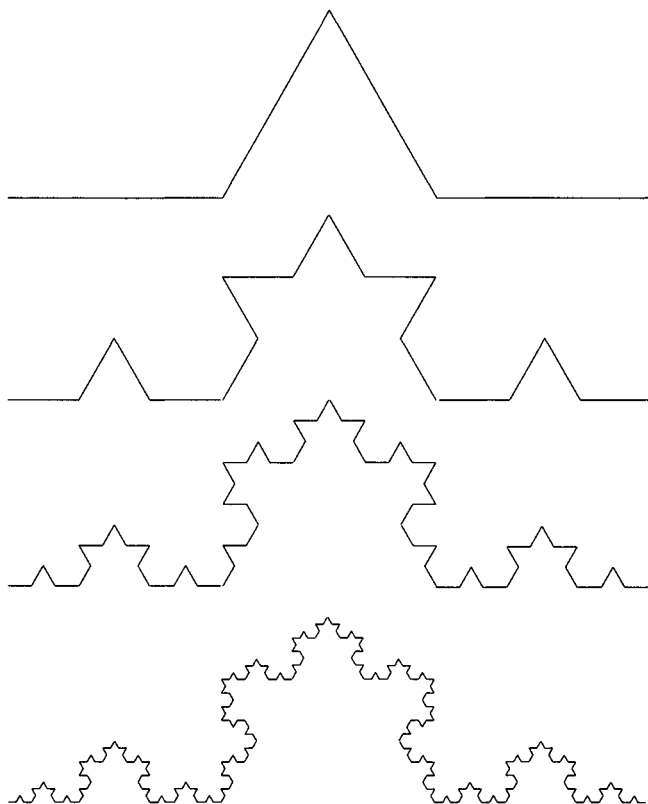


Figure 4.3. The von Koch curve.

compact subsets of $\mathbf{R}^n$ is a complete metric space with the *Hausdorff metric* ρ ,

$$\rho(E, F) = \max\{d(x, F), d(y, E) : x \in E, y \in F\},$$

see e.g. Federer [3, 2.10.21] or Rogers [1, §2.6]. The map $\tilde{S}: E \mapsto \bigcup_{i=1}^N S_i E$ is readily seen to be a contraction in the Hausdorff metric, whence it has a unique fixed point, cf. e.g. Edgar [1, 2.1.36]. By definition, this is the invariant set we wanted.

In addition, it follows by the simple general properties of contractions in complete metric spaces that however we choose an initial compact set $F \subset \mathbf{R}^n$, the iterations

$$\tilde{S}^m(F) = \tilde{S} \circ \cdots \circ \tilde{S}(F) = \bigcup_{i_1=1}^N \cdots \bigcup_{i_m=1}^N S_{i_1} \circ \cdots \circ S_{i_m}(F)$$

will converge to K . Moreover, for any m the set K satisfies

$$K = \bigcup_{i_1=1}^N \cdots \bigcup_{i_m=1}^N S_{i_1} \circ \cdots \circ S_{i_m}(K).$$

Since

$$d(S_{i_1} \circ \cdots \circ S_{i_m}(K)) \leq (\max_{1 \leq i \leq N} r_i)^m d(K) \rightarrow 0, \quad \text{as } m \rightarrow \infty,$$

an invariant set can be expressed as a union of arbitrarily small sets geometrically similar to itself.

We define an invariant set under $\mathcal{S}$ to be *self-similar* if with $s = \dim K$,

$$\mathcal{H}^s(S_i(K) \cap S_j(K)) = 0 \quad \text{for } i \neq j.$$

This definition is rather awkward to use, but the following somewhat stronger separation condition, called the *open set condition*, is very convenient:

There is a non-empty open set O such that

$$\bigcup_{i=1}^N S_i(O) \subset O \quad \text{and} \quad S_i(O) \cap S_j(O) = \emptyset \quad \text{for } i \neq j.$$

This is satisfied if the different parts $S_i(K)$ are disjoint as for the classical Cantor sets. Then we can use as O the ε -neighbourhood $\{x : d(x, K) < \varepsilon\}$ for sufficiently small ε . The open set condition also holds in many other interesting cases. For example, in the case of the von Koch curve we can take for O the open triangle which is the interior of the convex hull of the polygonal line consisting of the first four line segments, see Figure 4.3. Under the open set condition the dimension of K is explicitly determined by the contraction ratios $r_1, \dots, r_N$ of the similitudes S_i in $\mathcal{S}$:

4.14. Theorem. *If $\mathcal{S}$ satisfies the open set condition, then the invariant set K is self-similar and $0 < \mathcal{H}^s(K) < \infty$, whence $s = \dim K$, where s is the unique number for which*

$$(1) \quad \sum_{i=1}^N r_i^s = 1.$$

Moreover, there are positive and finite numbers a and b such that

$$ar^s \leq \mathcal{H}^s(K \cap B(x, r)) \leq br^s \quad \text{for } x \in K, 0 < r \leq 1.$$

For a proof see Hutchinson [1] or Falconer [4].

If in the above $r_1 = \cdots = r_N = r$ we have $\dim K = \log N / \log(1/r)$ in accordance with what we previously proved about the Cantor sets $C(\lambda)$. For the von Koch curve K this gives $\dim K = \log 4 / \log 3$.

4.15. Further remarks on self-similarity and related concepts.

Theorem 4.14 was essentially proven by Moran [1] as long ago as the forties. But he did not take the sequences of similitudes as the generating objects. This point of view of Hutchinson is extremely useful for developing the theory and for the computer generation of pictures of self-similar sets. Hutchinson's paper also illuminated considerably the constructions in the book of Mandelbrot [1]. After Hutchinson's work a huge literature on self-similar and related sets and measures has grown. The references below cover only part of it. For general views, see the books Bandt, Flachsmeier and Haase [1], Barnsley [1], Barnsley and Demko [1], Bélair and Dubuc [1], Edgar [1], Falconer [4], [16], Guzmán, Martin, Morán and Reyes [1], Mandelbrot [1], Peitgen and Richter [1] and Peitgen and Saupe [1]. We make a few more specific comments on the recent literature.

The open set condition has been used and studied in many papers. Schief [1] showed that it is equivalent to the apparently stronger form where the open set O is required to intersect the self-similar set K . Moreover, he showed that these conditions hold if and only if the invariant set K has positive and finite $\mathcal{H}^s$ measure where s is given by 4.14 (1), see also Schief [2]. A characterization of the condition $0 < \mathcal{H}^s(K) < \infty$ was also given by Bandt and Graf [1]. Falconer [11] proved for some "generic" sequences of similitudes that the Hausdorff dimension of the invariant set is given as in Theorem 4.14 without the assumption about the open set condition. We shall give this proof in 9.13. Strichartz [2] and [5]–[7] studied the behaviour of Fourier transforms of measures on self-similar and related sets, see also Lau [2], Lau and Wang [1] and Jorgensen and Pedersen [1]. Other results on self-similar sets and closely related constructions can be found e.g. in Andersson [1], Bandt [1]–[3], Bandt and Kuschel [1], Bandt and Stahnke [1], Barlow and Bass [1], Barlow and Perkins [1], Barnsley [2], Brucks [1], Dekking [1]–[2], Deliu, Geronimo, Shonkwiler and Hardin [1], Falconer [15], [25], Fan [1], Feiste [1], Gadde [1], Geronimo and Hardin [1], Graf [2], Hata [1], [4], Hayashi [1], Kenyon [1], Kigami [1], Lalley [1], Lindstrøm [1], Marion [2]–[3], Mattila [5], Mauldin and Williams [3], Morán [1]–[4], Peres [2], Reyes [1]–[2], Rushing [1], Spear [1], Stella [1], Tsujii [1], Wallin [1], Wallin and Wingren [1] and Wicks [1].

Instead of keeping the sequence of similitudes fixed at each stage of the construction, one could let it vary randomly according to a given probability distribution. Such *statistically self-similar* sets have been studied e.g. in Arbeiter [1], Falconer [9], [14], [16, Chapter 15], Falconer and Grimmett [1], Graf [1], Graf, Mauldin and Williams [1], Hawkes [3]–

[4], Mauldin and Williams [1] and Peyrière [2]. In these works one often finds a formula which gives almost surely the Hausdorff dimension. A more precise result was derived by Graf, Mauldin and Williams [1]. They were able to find the dimension function h for which the random self-similar set has almost surely positive and finite Λ_h measure. A closely related theory was developed by U. Zähle [1]–[3], see also Patzschke and U. Zähle [1], and Patzschke and M. Zähle [2] and [4].

If the similitudes are replaced by general contractions the invariant compact set can be shown to exist as above. The case where these contractions are affine maps $\mathbf{R}^n \rightarrow \mathbf{R}^n$ has been widely studied. The computation of the Hausdorff dimension of such *self-affine* sets turns out to be much more difficult. The first such results were found by Bedford [1] and McMullen [1]; for extensions and refinements, see Bedford and Urbanski [1], Gatzouras and Lalley [1], Kenyon and Peres [2], Lalley and Gatzouras [1] and Peres [1]. Falconer [12] derived a dimension formula for “generic” self-affine sets in terms of the singular values of the maps, see also Falconer and Marsh [1], and estimated the dimension in the same spirit for a fixed self-affine set in Falconer [18], see also Falconer [16, 9.4].

Many graphs of nowhere differentiable functions are self-affine in the above sense or in a somewhat modified sense. One of the intriguing open questions is whether the Hausdorff dimension of the graph of the Weierstrass function

$$\sum_{k=1}^{\infty} \lambda^{(s-2)k} \sin(\lambda^k t), \quad t \in \mathbf{R},$$

where $\lambda > 1$, $1 < s < 2$, equals s . A good estimate has been given by Mauldin and Williams [2]. For other results, see e.g. Barnsley, Elton, Hardin and Massopust [1], Bedford [3]–[4], Chamizo and Córdoba [1], Dubuc [1], Dubuc and Tricot [1], Edgar [1], Falconer [16], Hata [3], Hu and Lau [1]–[3], Kamae [1], Kono [1], Ledrappier [1], Przytycki and Urbanski [1], Rezakhanlou [1], Shiota and Sekiguchi [1] and Urbanski [1]–[2].

Limit sets of Möbius groups

4.16. The book of Nicholls [1] is a good source for the topics described below. Let G be an infinite discrete group of Möbius transformations acting on the unit ball $B(1)$ in $\mathbf{R}^n$, $n \geq 2$. Then G is countable and for any x in the open ball $U(1)$ the images gx , $g \in G$, accumulate only on

the boundary S^{n-1} . The set $L(G)$ of the limit points of $\{gx : g \in G\}$ is called the limit set of G . It is a compact subset of S^{n-1} independent of x and for many groups G it is a fractal. Often one finds the Hausdorff dimension of $L(G)$ by studying the series

$$\sum_{g \in G} e^{-\delta d(0, g0)}, \quad \delta > 0,$$

where d is the hyperbolic metric of $U(1)$. Let $\delta(G)$ be the infimum of the values $\delta > 0$ for which this series converges. Then $\delta(G) \leq n - 1$ and for example for the so-called geometrically finite groups G ,

$$\delta(G) = \dim L(G) < n - 1$$

by the results of Patterson [1], Sullivan [3] and Tukia [1], see also Stratmann and Velani [1] and Tukia [2]. Many interesting measure-theoretic and other results were derived by Bishop and Jones [1].

Such results as described above can be proven and more insight found with the help of G -conformal measures μ of dimension δ on $L(G)$ which are characterized by the property

$$\mu(gA) = \int_A |g'|^\delta d\mu \quad \text{for } g \in G$$

and for Borel sets $A \subset L(G)$. Here $|g'|$ is the norm of the derivative g' . Relying on the earlier work of Patterson [1], D. Sullivan [1] introduced them in this form. Patterson and Sullivan showed that in many cases such a conformal measure exists for $\delta = \delta(G)$. It can be obtained as a weak limit of suitable linear combinations of the Dirac measures δ_{g0} , $g \in G$. Sometimes it is equivalent to $\mathcal{H}^{\delta(G)} \llcorner L(G)$ and sometimes to the packing measure on $L(G)$, which will be introduced in 5.10, see Sullivan [3].

In Velani [1] and Dodson, Melián, Pestana and Velani [1] the Patterson–Sullivan measures were used to study some subsets of $L(G)$ which have a number-theoretic (Diophantine approximation) flavour. Results on the Hausdorff dimension of various number-theoretic sets are described in Rogers [1, § 3.2], Falconer [4, 8.5] and [16, Chapter 10]; for more recent ones, see Baker [1], Dickinson [1], Dodson [1]–[2], Dodson and Hasan [1], Dodson, Rynne and Vickers [1]–[2] and Mauldin and Urbánski [2]. The last paper also deals with Apollonian packing. For more on this topic, see e.g. Falconer [4, 8.4] and Tricot [3], and in particular Brooks [1] for relations to discrete groups.

Dynamical systems and Julia sets

4.17. In dynamical systems one is often led to study an expanding map f and a closed invariant set J such that $f(J) = J = f^{-1}(J)$. In many cases J is a fractal and it has a nearly self-similar or self-affine structure, which can be a starting point for introducing measures on J related to the dynamics of f and for estimating the Hausdorff dimension of J . Often J is characterized as the largest set where the iteration $x \mapsto f(x) \rightarrow f(f(x)) \rightarrow \dots$ is in some sense chaotic. For general descriptions, see Falconer [4, 8.7] and [16, Chapters 13 and 14].

A widely investigated particular case is that of the Julia sets J of complex-analytic or meromorphic maps f in the complex plane $\mathbf{C}$. If $f(z) = z^2$, the Julia set is the unit circle S^1 . Adding a small constant $c \in \mathbf{C}$ the Julia set J_c of $z \mapsto z^2 + c$ is a fractal Jordan curve. For c further away from the origin J_c assumes various forms; it may be a self-intersecting fractal curve or a Cantor-type set. The Mandelbrot set M serves as a directory; the parameters c in different parts of M correspond to different properties of J_c . For pictures and explanations, see Peitgen and Richter [1]. Recent good books on the iteration of rational maps in $\mathbf{C}$ are those of Beardon [2] and Carleson and Gamelin [1].

For the polynomials $z^2 + c$ the Julia set J_c is close to S^1 when c is close to zero. Ruelle [1] proved for the Hausdorff dimension that $\dim J_c \approx 1 + |c|^2 / \log 16$ for small c and that $\dim J_c$ is a real-analytic function of $c \in \mathbf{C}$. He also gave a formula, see Ruelle [2], based on the earlier work of Bowen [1] for $\dim J_c$ in terms of the ergodic-theoretic concepts of pressure and entropy. In fact, Ruelle proved such results for much more general transformations $f: J \rightarrow J$. The works of Bowen and Ruelle have inspired many later investigations, see e.g. Bedford [1]–[2], Denker and Seck [1], Denker and Urbański [1]–[3], Falconer [13], [15], [21], Falconer and Marsh [2], McLaughlin [1], Mauldin and Urbański [1] and K. Simon [1]. Many of the aforementioned papers on self-similar and self-affine sets also contain dynamical aspects, and the literature on the measure- and dimension-theoretic properties of dynamical systems has exploded during the last ten years. A theory applying to a variety of situations has been developed by Mauldin and Urbański [2]. Dynamical systems related to Cantor sets have been studied for example in Sullivan [4], Bedford's article in Bélair and Dubuc [1], Bedford and Fisher [2]–[3] and Rand [1].

Another somewhat related approach to studying the geometric measure-theoretic properties of Julia sets was developed by Sullivan [2]. In analogy to the case of limit sets discussed above Sullivan showed that

if f is a rational map on $\mathbf{C}$, there exists for some positive numbers δ a Radon measure μ on the Julia set J of f such that

$$\mu(f(A)) = \int_A |f'(z)|^\delta d\mu z$$

for Borel sets $A \subset J$. Such f -conformal measures are obtained as weak limits of measures of the form $\sum_{f^k(y)=z} c(y) \delta_y$ as $k \rightarrow \infty$ where z is a point outside J and f^k is the k -th iterate of f . Suppose f is expanding on J , that is, there is k such that $|(f^k)'(z)| > 1$ for $z \in J$. Then, as Sullivan showed, for $\delta = \dim J$ there exists a unique f -conformal probability measure and it is comparable to $\mathcal{H}^\delta \llcorner J$. Later Denker and Urbański [1]–[4] have extended such results to many non-expanding cases. As in the case of limit sets the f -conformal measure turns out to be sometimes equivalent to the packing measure on J , see Denker and Urbański [2].

Recently Shishikura [1] has shown that the boundary of the Mandelbrot set has Hausdorff dimension two. Astala [1] used fractal type methods with dynamical systems to solve several problems in the theory of plane quasiconformal mappings, including a problem on Hausdorff dimension.

Harmonic measure

4.18. Let G be an open connected set in $\mathbf{R}^n$ with not too small boundary F (this means that F has positive classical capacity). For $x \in G$ let w_x be the harmonic measure at x . Roughly speaking this is to say that w_x is a Radon probability measure on F such that for bounded continuous functions $\varphi: F \rightarrow \mathbf{R}$ the harmonic function in G with boundary values φ is given by $x \mapsto \int \varphi dw_x$. The measure w_x can also be described as the first hitting probability on F for the Brownian motion starting from x . For this and other properties of the harmonic measure, see e.g. Garnett [3] and Pommerenke [1].

For any x_1 and x_2 in G the measures w_{x_1} and w_{x_2} are mutually absolutely continuous, thus we can speak about the absolute continuity and singularity of the harmonic measure on F with respect to Hausdorff measures without referring to any particular point x . Øksendal [1] proved that the harmonic measure and the Lebesgue measure $\mathcal{L}^n$ are always mutually singular. Note that $\mathcal{L}^n(F)$ may be positive but Øksendal's result tells us that there is $A \subset F$ such that $\mathcal{L}^n(A) = 0$ and $w_x(F \setminus A) = 0$. This is a reflection of the fact that the harmonic measure tends to concentrate on the parts of the boundary of G which are more “easily accessible”

from G . From the Brownian motion point of view this is quite natural; the most accessible points are most likely to be hit first. If the boundary is very complicated, most of it is “hidden” and only a small part easily accessible. Guided by this Øksendal [2] also conjectured that in $\mathbf{R}^2$ the harmonic measure is singular with respect to the Hausdorff measures $\mathcal{H}^s$ for $s > 1$. This was proved by Makarov [1] for simply connected domains $G \subset \mathbf{R}^2$. Moreover, he obtained much more precise information about relations between harmonic measure and the generalized Hausdorff measures Λ_h , see also Pommerenke [1]. Earlier Carleson [2] had proved such results for many self-similar, not necessarily connected, boundaries. He employed interesting dynamical methods. Later this aspect has been developed e.g. by Przytycki, Urbański and Zdunik [1], Volberg [1] and Denker and Urbański [4]. Jones and Wolff [1] solved the problem for general domains in $\mathbf{R}^2$ by showing that the harmonic measure always lives in a set of Hausdorff dimension at most one. Later Wolff [1] sharpened this by showing that it lives in a set of σ -finite $\mathcal{H}^1$ measure.

In $\mathbf{R}^n$ Bourgain [2] proved that there is a constant $w(n) < n$ such that the harmonic measures in $\mathbf{R}^n$ are singular with respect to the Hausdorff measures $\mathcal{H}^s$ for $s > w(n)$. According to the above $w(2) = 1$. Wolff [2] showed that $w(3) > 2$.

For other properties of harmonic measure mixing geometry and complex analysis, see e.g. Carleson and Jones [1] and Jones and Makarov [1].

Exercises.

1. Let U be an open ball in $\mathbf{R}^n$, $n \geq 2$, with $d(U) = \delta$. Show that for $0 \leq s \leq 1$, $\mathcal{H}_\delta^s(U) = \mathcal{H}_\delta^s(\overline{U}) = \mathcal{H}_\delta^s(\partial U)$.
2. Prove Theorem 4.4.
3. Prove Lemma 4.6.
4. Show that if h and k are non-decreasing non-negative functions on $[0, \infty)$ with value 0 at 0 and if $\lim_{r \downarrow 0} (h(r)/k(r)) = 0$, then $\Lambda_k(A) < \infty$ implies $\Lambda_h(A) = 0$.
5. Construct an example to show that the conclusion of the preceding exercise may fail if the assumption $\lim_{r \downarrow 0} (h(r)/k(r)) = 0$ is replaced by $\liminf_{r \downarrow 0} (h(r)/k(r)) = 0$.
6. Prove that the Cantor-type set of 4.12 has positive and finite $\mathcal{H}^s$ measure.
7. Show that for the self-similar set $K = \bigcup_{i=1}^N S_i K$ the open set condition is satisfied if the different parts $S_i K$ are disjoint.
8. Prove that the Hausdorff metric ϱ is a metric.
9. Show that the map $\tilde{S}$ in 4.13 is a contraction with respect to ϱ .

10. Let $K = \bigcup_{i=1}^N S_i K$ be the invariant set under the similitudes $S_i: \mathbf{R}^n \rightarrow \mathbf{R}^n$. Show that if $S_1 K, \dots, S_N K$ are disjoint and $0 < \mathcal{H}^s(K) < \infty$, then $\sum_{i=1}^N \text{Lip}(S_i)^s = 1$.